大疆多协议高频头

硬件设计文档汇编

DESIGN · PINOUT · TASKS · BOM

2026-09-24(含 ESP-NOW 模式设计)· Alius

1. 设计文档

大疆多协议高频头 — 设计文档

> **2026-09-24 实施审计：未完成，不能投板。** 最新实况和依据见 执行报告。 当前工作工程为「遥控器射频板」；以下历史"完成/干净/已接线"描述仅代表迁移阶段，不代表电路设计或当前 DRC 通过。 632 个物理引脚中，210 个联网、6 个有依据 NC、416 个未连接；已确认 BATTERY/BATTERY_5V 电源桥接。

当前确定的设计要求

- 2–6S 电池输入与 USB 供电；USB 单独供电时限功率。
- ELRS 天线口目标 1W；nRF24L01+ / A7105 各 500mW；ESP-NOW 不要求外置大功率。
- RTL8812AU 保留裸芯片。已取得参考图，但其 BOM/射频器件与来源尚未完全核定，未据此落图。
- 所有物理脚须正确联网或有证据的 NC；不允许以单脚标签网或批量 NC 代替电路。
- 本轮已落图并验证的变更只有 U3 19–24 外部 NC；六脚用于封装内 Flash。

对旧稿的关键纠正

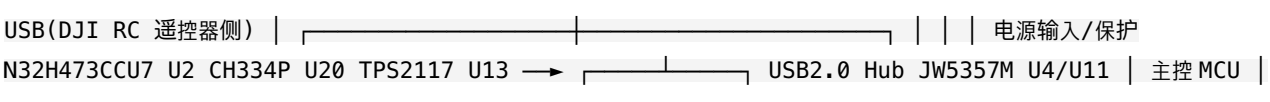
- U11 输入 BATTERY 与输出 BATTERY_5V 被 C18:2/C19:1 之间的共线导线桥接；TPS2117 VIN1/MODE 因此读为 BATTERY。需修复桥接并重设计高压降压。不能只改网名。
- U4 属 IN_5V→3V3 低压域，不因 6S 需求就一概更换；U11 必须按实际高压输入重新选型。
- CH334P 用 12MHz 晶振且无 RREF 脚；原任务表 24MHz/RREF 要求错误。
- DPX205850DT-4055A1 为射频双工器，不能用作 USB ESD/数据线滤波；USB1/2/3 尚未处置。
- ESP32-C3 microSD 应按 SPI 设计；GPIO18/19 为 USB D−/D+，不是默认 UART0。与主控现有 GPIO4/5 连接需配合固件映射。
- U15 实际物理封装与符号映射仍待核实；不能仅凭 5 脚符号推断完整 TCXO 接法。
- 图纸当前还有出界、标签重叠及孤立图元，器件本体不重叠不等于完整布局通过。

资料来源与适用边界见 REFERENCES.md。原稿、生成脚本已原样保存于 `implementation/2026-09-24/baseline/documents/`。

以下为历史架构与迁移记录；与上述实况冲突时以上述审计为准。

> DJI RC 配套高频头:多协议 2.4G 射频链 + RTL8812AU 数字 VTX,USB 接入 DJI RC 遥控器。
>
> EasyEDA 工程:`DJI RC Plus改造计划` / board「大疆多协议高频头」 原理图 **schematic9**(uuid `de9f23bc06e9e3b6`),四页:POWER / MCU / VTX / LoRa PCB **PCB8**(uuid `a6b755b4dd3530b7`,与 schematic9 linked) 器件总数:153(2026-09-23 实测)

1. 系统架构



```
│ (buck ×2) │ SWD/IO/串口│ ┌──────┴──────┐ │ └──────┬──────┘ │ │ │ SPI/控制总线 │ │ │
┌──────┴──────┴──────┴──────┴──────┘ │ │ │ │ │ RTL8812AU U21 │ U3 U5 U19 U27 (2T2R WiFi, │ ESP32-
SX1281 NRF24 A7105 5.8G VTX, │ C3FH4 (ELRS) (玩具) (FlySky) 40M 晶振 X3) │ │ │ │ │ │
└──────┬──────┬──────┬──────┘ │ │ SKY13414 U29 IPEX 天线 ×2 │ (SP4T 天线开关) (视频链路) │ │ │ AT2401C
U17(2.4G PA,nRF24 链路) │ │ └──────────── 2.4G 天线(控制链路)
```

- **控制链路(2.4G)**:四合一射频共享天线,SKY13414 SP4T 分时切换,单协议同一时刻在线。
- **视频链路(5.8G)**:RTL8812AU 经 CH334P Hub 挂在 USB 总线上,wifibroadcast 类数字 VTX/VRX。

2. 协议映射(四合一射频)

芯片	位号	主要协议/生态	典型用途
SX1281	U5(SX1281IMLTRT,52M TCXO U15)	**ExpressLRS 2.4G**	FPV 穿越机、长距离/低延迟 RC
nRF24L01	U19(NRF24L01P-R,QFN-20,16M 晶振 X6;PA=U17 AT2401C)	Bayang、Syma、Hisky、WLtoys、MJX、Cheerson、JJRC 等	国产玩具/微型无人机
A7105	U27(QFN-20,12M 晶振 X9)	FlySky AFHDS / AFHDS2A	富斯遥控器、接收机、国产航模
ESP32-C3FH4	U3(40M 晶振 X5,+microSD CN3)	ESP-NOW / Wi-Fi / BLE	DIY 遥控、机器人、智能车、物联网

四芯片 SPI 共总线(SPI_SCK/MISO/MOSI + 独立 NSS),由 N32H473 统一调度。

2.0 工程与页面映射(2026-09-24 干净重建完成)

工作工程:「遥控器射频板」——四页已全部干净重建,网表引擎/DRC 恢复:

项	UUID	状态
工程 遥控器射频板	`c15f2ba144e644eca844ab18ebaf753e`	工作工程
Board1(schematic1 ↔ PCB1)	sch `280977e0...` / pcb `3e4fbf69c50aec75`	绑定完好
POWER 页	`7620d079e8618c97`	23 件,**63/63 网已接线(引擎确认)**
MCU 页	`932d3fd21946e404`	49 件(含 USB Hub 组),**102/102 网已接线**
VTX 页	`364c5031568c17c2`	30 件(8812AU 图传组),待接线(WIP)
LoRa 页	`6da79bf818384203`	51 件(U19=NRF24L01P-R),**45/45 网已接线**

引擎恢复与干净重建(2026-09-24):复制内容携带几何冲突(引脚坐标重合/本体互压/页外) 使网表导出与 DRC 管线损坏。四页经"清空→防冲突放置(三审计)→幂等挂网"重建后, **导出恢复、netlist 视图恢复、DRC 四页 致命 0 / 错误 0**(64 警告=射频/VTX 链待接线引脚, 属设计进行中)。分工:**VTX=图传模块(RTL8812AU);USB Hub(CH334P+U14+U8/U9+R23)在 MCU 页**。 原 DJI RC Plus改造计划工程保留为历史。详见 migration/REPORT.md §9。

2.1 全部 IC/芯片清单(按图页,2026-09-24 更新)

POWER 页(4 颗)

位号	型号	厂商	作用	封装	C 号
U1	ZX-SH1.0-4PWT	MEGASTAR	电源/USB 输入连接器(4P 1.0mm 侧插):接入主机 USB 5V 供电与 USB_DP/DN 差分数据	CONN 4P-1.00	C7430446
U13	TPS2117DRLR	TI(德州仪器)	电源复用器 Power Mux(1.6—5.5V / 4A):USB 5V 与电池 5V 轨二选一自动切换(优先级可配),输出 IN_5V	SOT-583-8	C22399676
U4	JW5357MSOTB	Joulwatt(杰华特)	同步降压 DC-DC → **3V3 主数字轨**(绝对最大 20V,6S 输入需换型,见已知问题)	SOT-23-6	C5197556
U11	JW5357MSOTB	Joulwatt(杰华特)	同步降压 DC-DC → **BATTERY_5V 5V 轨**(供射频功放/VTX;同上耐压限制)	SOT-23-6	C5197556

MCU 页(10 颗)

位号	型号	厂商	作用	封装	C 号
U2	N32H473CCU7	Nations(国民技术)	**主控 MCU**:系统控制核心——SPI 主机调度四合一射频、UART 连 ESP32-C3、USB 通信、SWD 调试、IO 扩展与前端控制	QFN-48	C49191475
U20	CH334P	WCH(沁恒)	USB 2.0 Hub 控制器:上行接 U14 主机口,下行挂载 RTL8812AU 图传(其余口备用/可选 ESP32 USB)	QFN-16	C5373042
U14	ZX-SH1.0-4PWT	MEGASTAR	USB 上行连接器(主机侧数据+供电)	CONN 4P-1.00	C7430446
USB2	DPX205850DT-4055A1	TDK	集成 ESD 保护+滤波阵列:USB 差分对防护(器件侧)	滤波阵列	C2847617
USB3	DPX205850DT-4055A1	TDK	同上(连接器侧)	滤波阵列	C2847617
U8	RCLAMP0521T-ES	Nexperia(安世)	ESD 保护二极管对:USB_DP 静电防护	DFN1006-2L	C5180263
U9	RCLAMP0521T-ES	Nexperia(安世)	ESD 保护二极管对:USB_DN 静电防护	DFN1006-2L	C5180263
Q1	AO3401A	AOS(万代)	P 沟道 MOS(-30V/4A):电源闸门/防倒灌	SOT-23	C15127
X2	SX3B8.000F1010F30	星晶	**U2 主时钟晶振 8MHz**	3225 4P	C7302001
X1	SX3B8.000F1010F30	星晶	8MHz 晶振(疑为 X2 重复件,待裁决删除)	3225 4P	C7302001
X7	40MHz 3225	Generic	40MHz 晶振(疑为 RTL8812AU 时钟残留,与 X3 重复,待裁决)	3225 4P	C47089401
TP1	测试点	—	CHIP_EN 测试点	—	—

VTX 页(4 颗)

位号	型号	厂商	作用	封装	C 号
U21	RTL8812AU-CG	Realtek(瑞昱)	**数字图传核心**:2T2R 双频(2.4G/5.8G)WiFi USB 控制器——经 CH334P 挂 USB,wifibroadcast 类视频链;内部 SPS 需外置电感 L3+反馈 R18,两路 RF 独立双天线	QFN-76	C2928090
X3	40MHz 3225	Generic	RTL8812AU 时钟晶振	3225 4P	C47089401

位号	型号	厂商	作用	封装	C 号
Q3	AO3401A	AOS(万代)	VTX 电源闸门(V33_8812 轨前级,上电默认关)	SOT-23	C15127
TP2	测试点	—	VTX 调试测试点	—	—

LoRa 页(10 颗,四合一射频链)

位号	型号	厂商	作用	封装	C 号
U3	ESP32-C3FH4	Espressif(乐鑫)	**四合一 #4**:RISC-V SoC(内置 4MB flash),ESP-NOW / Wi-Fi / BLE 通信;兼 microSD 存储、与主控 UART 互联	QFN-32	C2858491
U5	SX1281IMLTRT	Semtech	**四合一 #1**:2.4G LoRa/FLRC/GFSK 收发器,ExpressLRS 2.4G 远距离低延迟控制链	QFN-24	C2151551
U19	NRF24L01P-R	NORDIC	**四合一 #2**:2.4G GFSK 收发器,玩具级多协议(Bayang/Syma/Hisky/WLtoys/MJX/Cheerson/JJRC 等)	QFN-20-EP	C8791
U27	A7105	AMICCOM(联杰)	**四合一 #3**:2.4G FSK 收发器,FlySky AFHDS / AFHDS2A 遥控协议	QFN-20	C126376
U29	SKY13414-485LF	Skyworks	SP4T 射频开关:四颗射频芯片分时共享一根 2.4G 天线的源选择(V1/V2/V3 控制)	QFN-14	C255353
U17	AT2401C	杭州中科微	2.4G 射频功放(+22dBm≈160mW):nRF24 链路 PA;1W 方案中作驱动级或移除(待选型)	QFN-16	C783588
U15	ABDFTCXO-52.000MHz	Abracon	52MHz TCXO 温补晶振:SX1281 高稳定时钟源	OSC 5P	C568572
USB1	DPX205850DT-4055A1	TDK	ESD 保护+滤波阵列(LoRa 页 USB 线防护)	滤波阵列	C2847617
Q2	AO3401A	AOS(万代)	RF 电源闸门(射频链供电,上电默认关)	SOT-23	C15127
X5	Q22FA12800101	NDK	ESP32-C3 时钟晶振 40MHz	2016 4P	C255898
X6	X322516MLB4SI	TYC(台湾晶技)	nRF24L01+ 时钟晶振 16MHz(CL=9pF)	3225 4P	C13738
X9	ABM8AIG-12.000MHZ	Abracon	A7105 时钟晶振 12MHz	3225 4P	C1986775
CN3	MUP-M627-1	HRO	microSD 卡座:ESP32 SDMMC 存储(飞行记录/配置)	MICRO-SD	C5205362

> 合计 28 颗(含连接器/晶振/测试点;纯 IC 19 颗)。四合一 = U3+U5+U19+U27,经 U29 共享天线, 遥控模式目标天线口 1W(前端选型中,TASKS T2)。逐步接线状态见 PINOUT.md。

2.2 U3 ESP-NOW 模式设计(参考 zvldz/ESP32-UART-Bridge 架构)

> 参考:zvldz/ESP32-UART-Bridge(MIT)。 该工程本身**无 ESP-NOW**(其无线链为 UDP/BLE/BT-SPP)且要求双核(不支持 C3)—— 我们借鉴其**桥接架构与交互范式**,在 U3(ESP32-C3FH4,单核)上实现 ESP-NOW 模式。

架构(UART ⇄ ESP-NOW 透传桥)

```
N32H473(U2) --USART1(C3TX/C3RX, 已接线)--> ESP32-C3(U3) │ 软件任务(FreeRTOS, 单核分时): ├─ uart_bridge:协议感知解析 │ · CRSF@420k(ELRS 帧格式,首选) │ · RAW 透传(按 3ms 空闲帧切包) │ · MAVLink v2(可选) ├─ espnow_link:收/发队列,重传+RSSI ├─ bind/led:绑定状态机 + 指示灯 └─ config:Web 配置页(AP 模式)/NVS 存储 ▼ ESP-NOW 对端(机载 ESP32 接收机/其它 peer)
```

与参考工程的映射与裁剪

参考工程(vzldz)	本设计(C3 裁剪版)
设备1/2/3 多 UART 路由	**仅 USART1 一路**(U2 ↔ U3,C3TX/C3RX 网已接)
UDP 多目标(≤4 IP)	**ESP-NOW 多 peer(≤4 MAC,NVS 存储)**;广播发现
BLE NUS / BT SPP	不做(与 ESP-NOW 模式互斥,C3 无 BT Classic)
CRSF 解析@420k 8N1	**保留**(与 ELRS 链一致的通道/链路质量帧)
RAW/MAVLink/SBUS 模式	RAW/MAVLink 保留;SBUS 不需要(有独立 RF 链)
Web 配置 + OTA	**保留**(AP 模式配置页:波特率/协议/peer 管理;OTA 备用)
GPIO0 BOOT 三击切换模式	**采纳**:绑定/模式切换交互复用 BOOT(GPIO9)或独立按键
GPIO21/48 RGB LED	**需新增 1 颗状态 LED**(C3FH4 无内置 RGB):连接/绑定/数据流指示

绑定与状态机(设计要点)

- **绑定**:长按 BOOT 3s(或配置页"添加 peer")进入 60s 绑定窗口,对端同理 → 交换 MAC 存 NVS; 支持定点(1 对 1)与广播(1 对多)两种 peer 模式。
- **LED 语义**(参考工程配色习惯):慢闪=等待绑定;常亮=已连接;快闪=数据流;双闪=配置模式。
- **失败回退**:ESP-NOW 连续丢包 >N → LED 快闪告警,U2 可经 UART 查询链路状态帧。

硬件需求与变更(并入 TASKS T8)

1. **新增器件(候选)**:绑定/模式按键 ×1(SW-SMD,建议接 GPIO0,10k 上拉)+ 状态 LED ×1(+1k 电阻, 建议空闲 GPIO,如 GPIO3)→ 走器件变更清单。
2. 已满足:USART1(C3TX/C3RX)、40MHz 时钟(X5)、供电 3V3、BOOT(GPIO9)strapping 上拉。
3. 固件基线:PlatformIO,ESP-IDF/Arduino esp-now API;参考工程的 CRSF 解析与 Web 配置代码 (MIT) 可移植,任务布局按单核分时改写(其双核要求是任务布局约束,非硬性)。

3. 电源树

```
电源输入 U1(SH1.0 4P) │ ├─ TPS2117 U13(电源自动切换 mux) │ ├─► JW5357M U4(buck) ──► 主数字轨 │ └─► JW5357M U11(buck) ──► 射频/VTX 轨 │ (L9/L10 APS252012M3R3F,22µF×4 CGA0805…) └─ Q2/Q3(AO3401A 电源闸门,按链路使能)
```

跨页电源网(保留原网名,不合并相近名):见 §6。

4. 时钟树

时钟	位号	服务对象
52.000MHz TCXO	U15(ABDFTCXO…-L-2-T5)	SX1281 U5
16MHz 晶振	X6(X322516MLB4SI)	NRF24L01 U19
12MHz 晶振	X9(ABM8AIG-12.000MHZ)	A7105 U27
40MHz 晶振	X5(Q22FA12800101)	ESP32-C3 U3
40MHz 晶振	X3(40MHZ/3225/12PF)	RTL8812AU U21
40MHz 晶振	X7	N32H473 U2(高速时钟)
8MHz 晶振	X1/X2(SX3B8.000…)	N32H473 U2(低速时钟)

5. 原理图页组织(schematic9 四页,2026-09-23 迁移)

页	uuid	内容	器件数	状态
POWER	`0c2c6a6f3ebfecc3`	电源输入、保护、稳压(power-io.P1 整页迁入)	23	**完成**:62/62 引脚网络
MCU	`e69fa11bb2c66323`	N32H473、IO、SWD(mcu.P9 + staging.P8 的 U2 组)	44	**完成**:92/92 引脚网络
VTX	`0d43196f0cf83159`	RTL8812AU 及外围(vtx.P6 + staging.P8 的 U21 组)	35	**完成**:10/10 引脚网络
LoRa	`46c48981caa3c9f9`	SX1281、nRF24L01、A7105、ESP32-C3 及 RF 前端(P2+P3+P4+P5)	51	**完成**:45/45 引脚网络

> 页名 LoRa 为用户指定名,实际承载全部四合一射频。8 个旧页已删除(全部快照于 `migration/snapshots/`); 连接基准为 PCB8 焊盘网表(原理图旧页从未成网);PCB 匿名网 `$1Nxx` 在原理图侧更名为 `PCB_1Nxx`。 执行细节、工具 bug 与恢复步骤见 `migration/REPORT.md`。

6. 跨页网络(迁移保持约束)

原 21 条跨页网(源自 Schematic8 单页读取记录,迁移前后逐引脚对账): `3V3` `GND` `SPI_SCK/MISO/MOSI` `SPI_NSS1/2` `RST1/2` `BUSY1/2` `IRQ1/2` `USART1_TX/RX` `USB_DP/DN/5V` `1.0_5V` `IN_5V` `XTA`

约束: – 电源网族 `BATTERY_5V / IN_5V / 5V / 1.0_5V / USB_5V` **各自独立命名,不因相近合并**; – 匿名网(`$1N…`)迁移后可能重编号,按**端点集合(ref.pin)等价**验收; – 保留全部 NC 标识与器件 uniqueId、位号、型号、封装及 PCB8 对应关系。

7. 沿革

1. **Schematic7**(`hardware/Netlist_Schematic7_2026-09-08.tel`):DJI RC Plus 内置射频模块逆向网表, N32H473 + ESP32-C3 + **双 LR1121**(旧方案)+ MXD8721 开关 + AT2401C PA。
2. **Schematic8**:单页整板(117 位号/488 引脚/124 网络),读取与七页拆分方案见 外部记录(Codex 2026-09-22 outputs)。
3. **Schematic9(现行)**:双 LR1121 方案废弃,替换为**四合一射频**(SX1281/nRF24L01/A7105/ESP32-C3), RF 开关换 SKY13414,新增 RTL8812AU VTX + CH334P Hub;由 8 个旧页整理为 POWER/MCU/VTX/LoRa 四页。

8. 已知问题清单

#	问题	影响	处置建议
1	~~U19 = NRF24L01(C1558392,无货)~~ **已于 2026-09-23 换绑为 NRF24L01P-R(C8791,NORDIC,QFN-20-EP,引脚兼容,原位姿)**,多协议固件 250kbps 档玩具协议所需的 + 版	已解决	—
2	**U11(JW5357M)原标注 2-6S(25.2V)超数据手册绝对最大 20V**	过压损坏风险	复核输入范围标注/分压,或换耐压更高的 buck
3	36 个无网络引脚中 8 个未按手册判定 NC	悬空脚定义不完整	逐脚按数据手册补 NC 或接线

#	问题	影响	处置建议
4	staging.P8 曾混装 U2+U21(主控与 VTX 同页)	已随四页迁移消除	—

9. 引脚定义与任务表

- PINOUT.md:35 颗芯片逐引脚定义(编号/名称/类型/当前网/设计建议)+ 悬空统计。
- TASKS.md:结合 PINOUT 悬空数据的接线任务分解(T0–T10,含依赖/阻塞/验收与执行顺序)。

完整逐引脚定义(编号/名称/类型/当前网/设计建议)见 PINOUT.md—— 按四页 35 颗芯片分表,含悬空脚统计(总览表)与接线路由草案,是接线设计的工作底稿。

10. BOM

见 BOM.md(153 器件,含立创 C 号与四页归属)。

2. 芯片引脚定义

全器件物理引脚审计

> 状态：未完成。联网仅表示网表成员关系，不表示电路正确；POWER 电池输入与降压输出存在已证实桥接。

工程：遥控器射频板；153 件，632 个物理引脚。

统计：connected=210；unconnected=416；nc=6。

来源：保留的 typed 快照，采集时间与 SHA-256 见 pin-audit.json。 重新采集后须更新 current-snapshots.json，再运行生成器；运行脚本本身不会刷新 EDA。

设计文档 · 任务表 · 本轮执行报告

POWER

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C15.1	1	PCB_1N6	connected	—
C15.2	2	PCB_1N14	connected	—
C18.1	1	GND	connected	—
C18.2	2	BATTERY	connected	—
C19.1	1	BATTERY	connected	—
C19.2	2	GND	connected	—
C20.1	1	3V3	connected	—
C20.2	2	PCB_1N9	connected	—
C27.1	1	BATTERY	connected	—
C27.2	2	PCB_1N10	connected	—
C29.1	1	GND	connected	—
C29.2	2	IN_5V	connected	—
C30.1	1	3V3	connected	—
C30.2	2	GND	connected	—
C9.1	1	PCB_1N5	connected	—
C9.2	2	PCB_1N13	connected	—
L10.1	1	PCB_1N6	connected	—
L10.2	2	BATTERY	connected	—
L9.1	1	PCB_1N5	connected	—
L9.2	2	3V3	connected	—
R10.1	1	BATTERY	connected	—
R10.2	2	PCB_1N10	connected	—
R11.1	1	PCB_1N94	connected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
R11.2	2	GND	connected	—
R12.1	1	PCB_1N94	connected	—
R12.2	2	BATTERY	connected	—
R15.1	1	GND	connected	—
R15.2	2		unconnected	—
R3.1	1	1.0_5V	connected	—
R3.2	2	5V	connected	—
R4.1	1	USB_5V	connected	—
R4.2	2	5V	connected	—
R5.1	1	GND	connected	—
R5.2	2	PCB_1N9	connected	—
R7.1	1	3V3	connected	—
R7.2	2	PCB_1N9	connected	—
R9.1	1	GND	connected	—
R9.2	2	PCB_1N10	connected	—
U1.1	1	1.0_5V	connected	—
U1.2	2	USB_DN	connected	—
U1.3	3	USB_DP	connected	—
U1.4	4	GND	connected	—
U1.5	5	GND	connected	—
U1.6	6	GND	connected	—
U11.1	BST	PCB_1N14	connected	—
U11.2	GND	GND	connected	—
U11.3	FB	PCB_1N10	connected	—
U11.4	EN	BATTERY	connected	—
U11.5	VIN	BATTERY	connected	—
U11.6	SW	PCB_1N6	connected	—
U13.1	GND	GND	connected	—
U13.2	VOUT	IN_5V	connected	—
U13.3	VIN1	BATTERY	connected	—
U13.4	PR1	PCB_1N94	connected	—
U13.5	MODE	BATTERY	connected	—
U13.6	VIN2	5V	connected	—
U13.7	VOUT	IN_5V	connected	—
U13.8	ST	PCB_1N91	connected	—
U4.1	BST	PCB_1N13	connected	—
U4.2	GND	GND	connected	—
U4.3	FB	PCB_1N9	connected	—
U4.4	EN	IN_5V	connected	—
U4.5	VIN	IN_5V	connected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U4.6	SW	PCB_1N5	connected	—

MCU

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C1.1	1	GND	connected	—
C1.2	2	3V3	connected	—
C10.1	1	GND	connected	—
C10.2	2	3V3	connected	—
C12.1	1	GND	connected	—
C12.2	2	3V3	connected	—
C14.1	1	GND	connected	—
C14.2	2	OSC_OUT	connected	—
C16.1	1	3V3	connected	—
C16.2	2	GND	connected	—
C2.1	1	3V3	connected	—
C2.2	2	GND	connected	—
C26.1	1	GND	connected	—
C26.2	2	NRST	connected	—
C3.1	1	3V3	connected	—
C3.2	2	GND	connected	—
C4.1	1	3V3	connected	—
C4.2	2	GND	connected	—
C40.1	1	C3XTAL_40K_P	connected	—
C40.2	2	GND	connected	—
C41.1	1	3V3	connected	—
C41.2	2	GND	connected	—
C47.1	1		unconnected	—
C47.2	2		unconnected	—
C5.1	1	3V3	connected	—
C5.2	2	GND	connected	—
C52.1	1		unconnected	—
C52.2	2		unconnected	—
C55.1	1	GND	connected	—
C55.2	2	CHIP_EN	connected	—
C56.1	1		unconnected	—
C56.2	2		unconnected	—
C57.1	1		unconnected	—
C57.2	2		unconnected	—
C59.1	1		unconnected	—
C59.2	2		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C6.1	1	3V3	connected	—
C6.2	2	GND	connected	—
C60.1	1		unconnected	—
C60.2	2		unconnected	—
C7.1	1	3V3	connected	—
C7.2	2	GND	connected	—
C8.1	1	GND	connected	—
C8.2	2	OSC_IN	connected	—
CN2.1	1	SWCLK	connected	—
CN2.2	2	SWDIO	connected	—
CN2.3	3	3V3	connected	—
CN2.4	4	GND	connected	—
CN4.1	1		unconnected	—
CN4.2	2		unconnected	—
CN4.3	3		unconnected	—
CN4.4	4		unconnected	—
L21.1	1		unconnected	—
L21.2	2		unconnected	—
L22.1	1		unconnected	—
L22.2	2		unconnected	—
Q1.1	G		unconnected	—
Q1.2	S		unconnected	—
Q1.3	D		unconnected	—
R1.1	1	CHIP_EN	connected	—
R1.2	2	3V3	connected	—
R14.1	1		unconnected	—
R14.2	2		unconnected	—
R16.1	1		unconnected	—
R16.2	2		unconnected	—
R17.1	1		unconnected	—
R17.2	2		unconnected	—
R19.1	1		unconnected	—
R19.2	2		unconnected	—
R2.1	1		unconnected	—
R2.2	2		unconnected	—
R20.1	1		unconnected	—
R20.2	2		unconnected	—
R22.1	1		unconnected	—
R22.2	2		unconnected	—
R23.1	1		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
R23.2	2		unconnected	—
R6.1	1	PCB_1N20	connected	—
R6.2	2	GND	connected	—
R8.1	1		unconnected	—
R8.2	2		unconnected	—
TP1.1	1	CHIP_EN	connected	—
U14.1	1	USB_5V	connected	—
U14.2	2	USB_DN	connected	—
U14.3	3	USB_DP	connected	—
U14.4	4	GND	connected	—
U14.5	5	GND	connected	—
U14.6	6	GND	connected	—
U2.1	VBAT	3V3	connected	—
U2.2	PC13	PC13	connected	—
U2.3	PC14–OSC32_IN		unconnected	—
U2.4	PC15–OSC32_OUT		unconnected	—
U2.5	PH0–OSC_IN	OSC_IN	connected	—
U2.6	PH1–OSC_OUT	OSC_OUT	connected	—
U2.7	PH6–NRST	NRST	connected	—
U2.8	PA0	PA0	connected	—
U2.9	PA1	PA1	connected	—
U2.10	PA2	PA2	connected	—
U2.11	PA3	PA3	connected	—
U2.12	PA4	PA4	connected	—
U2.13	PA5	PA5	connected	—
U2.14	PA6	PA6	connected	—
U2.15	PA7	PA7	connected	—
U2.16	PC4	PC4	connected	—
U2.17	PB0	BUSY1	connected	—
U2.18	PB1	BUSY2	connected	—
U2.19	PB2	RST1	connected	—
U2.20	VREF+	3V3	connected	—
U2.21	VDDA	3V3	connected	—
U2.22	PB10	V2–2	connected	—
U2.23	VDD	3V3	connected	—
U2.24	PB11	PB11	connected	—
U2.25	PB12	PB12	connected	—
U2.26	PB13	SPI_SCK	connected	—
U2.27	PB14	SPI_MISO	connected	—
U2.28	PB15	SPI_MOSI	connected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U2.29	PC6	PC6	connected	—
U2.30	PA8	SPI_NSS1	connected	—
U2.31	PA9	USART1_TX	connected	—
U2.32	PA10	USART1_RX	connected	—
U2.33	PA11	USB_DN	connected	—
U2.34	PA12	USB_DP	connected	—
U2.35	VDD	3V3	connected	—
U2.36	PA13	SWDIO	connected	—
U2.37	PA14	SWCLK	connected	—
U2.38	PA15	SPI_NSS2	connected	—
U2.39	PC10	PC10	connected	—
U2.40	PC11	PC11	connected	—
U2.41	PB3	RST2	connected	—
U2.42	PB4	IRQ1	connected	—
U2.43	PB5	IRQ2	connected	—
U2.44	PB6	V1–1	connected	—
U2.45	PB7	V2–1	connected	—
U2.46	PB8–BOOT0	PCB_1N20	connected	—
U2.47	PB9	V1–2	connected	—
U2.48	VDD	3V3	connected	—
U2.49	VSS	GND	connected	—
U20.1	XO		unconnected	—
U20.2	XI		unconnected	—
U20.3	DM4		unconnected	—
U20.4	DP4		unconnected	—
U20.5	DM3		unconnected	—
U20.6	DP3		unconnected	—
U20.7	DM2		unconnected	—
U20.8	DP2		unconnected	—
U20.9	DM1		unconnected	—
U20.10	DP1		unconnected	—
U20.11	DMU		unconnected	—
U20.12	DPU		unconnected	—
U20.13	RESET#/CDP		unconnected	—
U20.14	PGANG		unconnected	—
U20.15	V5		unconnected	—
U20.16	VDD33		unconnected	—
U20.17	GND		unconnected	—
U8.1	1	GND	connected	—
U8.2	2	USB_DN	connected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U9.1	1	GND	connected	—
U9.2	2	USB_DP	connected	—
USB2.1	GND		unconnected	—
USB2.2	COM		unconnected	—
USB2.3	GND		unconnected	—
USB2.4	LB		unconnected	—
USB2.5	GND		unconnected	—
USB2.6	HB		unconnected	—
USB3.1	GND		unconnected	—
USB3.2	COM		unconnected	—
USB3.3	GND		unconnected	—
USB3.4	LB		unconnected	—
USB3.5	GND		unconnected	—
USB3.6	HB		unconnected	—
X1.1	1		unconnected	—
X1.2	GND		unconnected	—
X1.3	3		unconnected	—
X1.4	GND		unconnected	—
X2.1	1	OSC_IN	connected	—
X2.2	GND	GND	connected	—
X2.3	3	OSC_OUT	connected	—
X2.4	GND	GND	connected	—
X7.1	1		unconnected	—
X7.2	GND		unconnected	—
X7.3	3		unconnected	—
X7.4	GND		unconnected	—

VTX

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C17.1	1		unconnected	—
C17.2	2		unconnected	—
C21.1	1		unconnected	—
C21.2	2		unconnected	—
C22.1	1		unconnected	—
C22.2	2		unconnected	—
C23.1	1		unconnected	—
C23.2	2		unconnected	—
C24.1	1		unconnected	—
C24.2	2		unconnected	—
C25.1	1		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C25.2	2		unconnected	—
C28.1	1		unconnected	—
C28.2	2		unconnected	—
C34.1	1		unconnected	—
C34.2	2		unconnected	—
C37.1	1		unconnected	—
C37.2	2		unconnected	—
C38.1	1		unconnected	—
C38.2	2		unconnected	—
C39.1	1		unconnected	—
C39.2	2		unconnected	—
C42.1	1		unconnected	—
C42.2	2		unconnected	—
C43.1	1		unconnected	—
C43.2	2		unconnected	—
C44.1	1		unconnected	—
C44.2	2		unconnected	—
C45.1	1		unconnected	—
C45.2	2		unconnected	—
C46.1	1		unconnected	—
C46.2	2		unconnected	—
C48.1	1		unconnected	—
C48.2	2		unconnected	—
C50.1	1		unconnected	—
C50.2	2		unconnected	—
C53.1	1		unconnected	—
C53.2	2		unconnected	—
C54.1	1		unconnected	—
C54.2	2		unconnected	—
C66.1	1		unconnected	—
C66.2	2		unconnected	—
C69.1	1		unconnected	—
C69.2	2		unconnected	—
L1.1	1		unconnected	—
L1.2	2		unconnected	—
L3.1	1		unconnected	—
L3.2	2		unconnected	—
Q3.1	G		unconnected	—
Q3.2	S		unconnected	—
Q3.3	D		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
R18.1	1		unconnected	—
R18.2	2		unconnected	—
R34.1	1		unconnected	—
R34.2	2		unconnected	—
TP2.1	1		unconnected	—
U21.1	V33_SYN		unconnected	—
U21.2	V12_SYN		unconnected	—
U21.3	V12_S0		unconnected	—
U21.4	RF2IN_S0		unconnected	—
U21.5	RF2IP_S0		unconnected	—
U21.6	V33_PAD2G_S0		unconnected	—
U21.7	RF2ON_S0		unconnected	—
U21.8	RF2OP_S0		unconnected	—
U21.9	V33_PAD5G_S0		unconnected	—
U21.10	RF5OP_S0		unconnected	—
U21.11	VD33PA5G_S0		unconnected	—
U21.12	RF5IP_S0		unconnected	—
U21.13	RF5IN_S0		unconnected	—
U21.14	V12_BB_S0		unconnected	—
U21.15	LNAON_S0		unconnected	—
U21.16	PAPE_2G_S0		unconnected	—
U21.17	PAPE_5G_S0		unconnected	—
U21.18	TRSW_S0		unconnected	—
U21.19	TRSWB_S0		unconnected	—
U21.20	VD12A		unconnected	—
U21.21	VD33A		unconnected	—
U21.22	XI		unconnected	—
U21.23	XO		unconnected	—
U21.24	CK_BT		unconnected	—
U21.25	ANTSW		unconnected	—
U21.26	ANTSWB		unconnected	—
U21.27	VD12D		unconnected	—
U21.28	VD33D		unconnected	—
U21.29	GPIO[4]/WL_ACT		unconnected	—
U21.30	GPIO[5]/BT_FREQ/BT_EN		unconnected	—
U21.31	GPIO[6]/BT_STATE		unconnected	—
U21.32	GPIO[7]/BT_PRI		unconnected	—
U21.33	LED2/GPIO[8]		unconnected	—
U21.34	LED1/GPIO[9]		unconnected	—
U21.35	LED0/GPIO[10]		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U21.36	GPIO[11]/PDn		unconnected	—
U21.37	GPIO[0]/BT_Suspend		unconnected	—
U21.38	GPIO[1]		unconnected	—
U21.39	GPIO[2]		unconnected	—
U21.40	GPIO[3]		unconnected	—
U21.41	GND		unconnected	—
U21.42	HSDP		unconnected	—
U21.43	HSDM		unconnected	—
U21.44	VD12S		unconnected	—
U21.45	VD33S		unconnected	—
U21.46	RREF		unconnected	—
U21.47	VDDTX		unconnected	—
U21.48	HSON		unconnected	—
U21.49	HSOP		unconnected	—
U21.50	GND		unconnected	—
U21.51	HSIN		unconnected	—
U21.52	HSIP		unconnected	—
U21.53	VDDRX		unconnected	—
U21.54	GND		unconnected	—
U21.55	LX_SPS0		unconnected	—
U21.56	LX_SPS0		unconnected	—
U21.57	VDSPS33		unconnected	—
U21.58	FB_SPS		unconnected	—
U21.59	VD33D		unconnected	—
U21.60	VD12D		unconnected	—
U21.61	TRSWB_S1		unconnected	—
U21.62	TRSW_S1		unconnected	—
U21.63	PAPE_5G_S1		unconnected	—
U21.64	PAPE_2G_S1		unconnected	—
U21.65	LNAON_S1		unconnected	—
U21.66	V12_BB_S1		unconnected	—
U21.67	RF2IN_S1		unconnected	—
U21.68	RF2IP_S1		unconnected	—
U21.69	V33_PAD2G_S1		unconnected	—
U21.70	RF2ON_S1		unconnected	—
U21.71	RF2OP_S1		unconnected	—
U21.72	V33_PAD5G_S1		unconnected	—
U21.73	RF5OP_S1		unconnected	—
U21.74	VD33PA5G_S1		unconnected	—
U21.75	RF5IP_S1		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U21.76	RF5IN_S1		unconnected	—
U21.77	GND		unconnected	—
X3.1	1		unconnected	—
X3.2	GND		unconnected	—
X3.3	3		unconnected	—
X3.4	GND		unconnected	—

LoRa

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C11.1	1	GND	connected	—
C11.2	2	3V3	connected	—
C13.1	1	GND	connected	—
C13.2	2	C3XTAL_40K_N	connected	—
C31.1	1		unconnected	—
C31.2	2		unconnected	—
C32.1	1		unconnected	—
C32.2	2		unconnected	—
C33.1	1		unconnected	—
C33.2	2		unconnected	—
C35.1	1		unconnected	—
C35.2	2		unconnected	—
C36.1	1	PCB_1N171	connected	—
C36.2	2	XTA	connected	—
C49.1	1		unconnected	—
C49.2	2		unconnected	—
C51.1	1		unconnected	—
C51.2	2		unconnected	—
C58.1	1		unconnected	—
C58.2	2		unconnected	—
C61.1	1		unconnected	—
C61.2	2		unconnected	—
C62.1	1		unconnected	—
C62.2	2		unconnected	—
C63.1	1		unconnected	—
C63.2	2		unconnected	—
C64.1	1		unconnected	—
C64.2	2		unconnected	—
C65.1	1		unconnected	—
C65.2	2		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
C67.1	1		unconnected	—
C67.2	2		unconnected	—
C68.1	1		unconnected	—
C68.2	2		unconnected	—
CN1.1	1	C3RX	connected	—
CN1.2	2	C3TX	connected	—
CN1.3	3	3V3	connected	—
CN1.4	4	GND	connected	—
CN3.0	0		unconnected	—
CN3.1	DAT2		unconnected	—
CN3.2	CD/DAT3		unconnected	—
CN3.3	CMD		unconnected	—
CN3.4	VDD		unconnected	—
CN3.5	CLK		unconnected	—
CN3.6	VSS		unconnected	—
CN3.7	DAT0		unconnected	—
CN3.8	DAT1		unconnected	—
CN3.9	CARDDETECT		unconnected	—
CN3.10	CARDDETECT		unconnected	—
L18.1	1		unconnected	—
L18.2	2		unconnected	—
L19.1	1		unconnected	—
L19.2	2		unconnected	—
L2.1	1	3V3	connected	—
L2.2	2	PCB_1N169	connected	—
L20.1	1		unconnected	—
L20.2	2		unconnected	—
L23.1	1		unconnected	—
L23.2	2		unconnected	—
L24.1	1		unconnected	—
L24.2	2		unconnected	—
L25.1	1		unconnected	—
L25.2	2		unconnected	—
L26.1	1		unconnected	—
L26.2	2		unconnected	—
L8.1	GND		unconnected	—
L8.2	OUT/IN		unconnected	—
L8.3	GND		unconnected	—
L8.4	IN/OUT		unconnected	—
Q2.1	G		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
Q2.2	S		unconnected	—
Q2.3	D		unconnected	—
R21.1	1		unconnected	—
R21.2	2		unconnected	—
R24.1	1		unconnected	—
R24.2	2		unconnected	—
R25.1	1		unconnected	—
R25.2	2		unconnected	—
R26.1	1		unconnected	—
R26.2	2		unconnected	—
R27.1	1		unconnected	—
R27.2	2		unconnected	—
R28.1	1		unconnected	—
R28.2	2		unconnected	—
R29.1	1		unconnected	—
R29.2	2		unconnected	—
R30.1	1		unconnected	—
R30.2	2		unconnected	—
R31.1	1		unconnected	—
R31.2	2		unconnected	—
R32.1	1		unconnected	—
R32.2	2		unconnected	—
R33.1	1		unconnected	—
R33.2	2		unconnected	—
U15.1	NC/VC	GND	connected	—
U15.2	GND	GND	connected	—
U15.3	OUT1	PCB_1N171	connected	—
U15.4	VDD	PCB_1N169	connected	—
U15.5	OUT2		unconnected	—
U17.1	GND		unconnected	—
U17.2	GND		unconnected	—
U17.3	GND		unconnected	—
U17.4	TXRX		unconnected	—
U17.5	TXEN		unconnected	—
U17.6	RXEN		unconnected	—
U17.7	GND		unconnected	—
U17.8	GND		unconnected	—
U17.9	GND		unconnected	—
U17.10	ANT		unconnected	—
U17.11	GND		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U17.12	GND		unconnected	—
U17.13	DNC		unconnected	—
U17.14	VDD		unconnected	—
U17.15	GND		unconnected	—
U17.16	VCC		unconnected	—
U17.17	EP		unconnected	—
U19.1	CE		unconnected	—
U19.2	CSN		unconnected	—
U19.3	SCK		unconnected	—
U19.4	MOSI		unconnected	—
U19.5	MISO		unconnected	—
U19.6	IRQ		unconnected	—
U19.7	VDD		unconnected	—
U19.8	VSS		unconnected	—
U19.9	XC2		unconnected	—
U19.10	XC1		unconnected	—
U19.11	VDD_PA		unconnected	—
U19.12	ANT1		unconnected	—
U19.13	ANT2		unconnected	—
U19.14	VSS		unconnected	—
U19.15	VDD		unconnected	—
U19.16	IREF		unconnected	—
U19.17	VSS		unconnected	—
U19.18	VDD		unconnected	—
U19.19	DVDD		unconnected	—
U19.20	VSS		unconnected	—
U19.21	EP		unconnected	—
U27.1	RSSI		unconnected	—
U27.2	BPBG		unconnected	—
U27.3	RFI		unconnected	—
U27.4	RFO		unconnected	—
U27.5	VDA2		unconnected	—
U27.6	VT		unconnected	—
U27.7	GND		unconnected	—
U27.8	XI		unconnected	—
U27.9	XO		unconnected	—
U27.10	VDA3		unconnected	—
U27.11	SCS		unconnected	—
U27.12	SCK		unconnected	—
U27.13	DVDD		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U27.14	SDIO		unconnected	—
U27.15	GND		unconnected	—
U27.16	GIO1		unconnected	—
U27.17	GIO2		unconnected	—
U27.18	CKO		unconnected	—
U27.19	REGI		unconnected	—
U27.20	VDA1		unconnected	—
U27.21	EXP		unconnected	—
U29.1	N/C		unconnected	—
U29.2	RF3		unconnected	—
U29.3	RF1		unconnected	—
U29.4	VDD		unconnected	—
U29.5	V3		unconnected	—
U29.6	V2		unconnected	—
U29.7	V1		unconnected	—
U29.8	N/C		unconnected	—
U29.9	RF2		unconnected	—
U29.10	RF4		unconnected	—
U29.11	N/C		unconnected	—
U29.12	N/C		unconnected	—
U29.13	ANT		unconnected	—
U29.14	N/C		unconnected	—
U29.15	EP		unconnected	—
U3.1	LNA_IN	PCB_1N60	connected	—
U3.2	VDD3P3	3V3	connected	—
U3.3	VDD3P3	3V3	connected	—
U3.4	XTAL_32K_P	C3GPIO0	connected	—
U3.5	XTAL_32K_N	C3GPIO1	connected	—
U3.6	GPIO2	C3GPIO2	connected	—
U3.7	CHIP_EN	CHIP_EN	connected	—
U3.8	GPIO3	C3GPIO3	connected	—
U3.9	MTMS	USART1_TX	connected	—
U3.10	MTDI	USART1_RX	connected	—
U3.11	VDD3P3_RTC	3V3	connected	—
U3.12	MTCK		unconnected	—
U3.13	MTDO		unconnected	—
U3.14	GPIO8	C3GPIO8	connected	—
U3.15	GPIO9	C3GPIO9	connected	—
U3.16	GPIO10	C3GPIO10	connected	—
U3.17	VDD3P3_CPU	3V3	connected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U3.18	VDD_SPI	3V3	connected	—
U3.19	SPIHD		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.20	SPIWP		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.21	SPICS0		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.22	SPICLK		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.23	SPID		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.24	SPIQ		nc	ESP32-C3 Datasheet v2.4 §2.6/Table 2-12: in-package Flash pins must not be used for other purposes
U3.25	GPIO18	C3GPIO18	connected	—
U3.26	GPIO19	C3GPIO19	connected	—
U3.27	U0RXD	C3RX	connected	—
U3.28	U0TXD	C3TX	connected	—
U3.29	XTAL_N	C3XTAL_40K_N	connected	—
U3.30	XTAL_P	C3XTAL_40K_P	connected	—
U3.31	VDDA	3V3	connected	—
U3.32	VDDA	3V3	connected	—
U3.33	EP	GND	connected	—
U5.1	VR_PA		unconnected	—
U5.2	VDD_IN		unconnected	—
U5.3	NRESET		unconnected	—
U5.4	XTA		unconnected	—
U5.5	GND		unconnected	—
U5.6	XTB		unconnected	—
U5.7	BUSY		unconnected	—
U5.8	DIO1		unconnected	—
U5.9	DIO2		unconnected	—
U5.10	DIO3		unconnected	—
U5.11	VBAT_IO		unconnected	—
U5.12	DCC_FB		unconnected	—
U5.13	GND		unconnected	—
U5.14	DCC_SW		unconnected	—
U5.15	VBAT		unconnected	—
U5.16	MISO_TX		unconnected	—
U5.17	MOSI_RX		unconnected	—
U5.18	SCK_RTSN		unconnected	—
U5.19	NSS_CTS		unconnected	—

位号.物理脚	符号脚名	实际网络	状态	NC 依据
U5.20	GND		unconnected	—
U5.21	GND		unconnected	—
U5.22	RFIO		unconnected	—
U5.23	GND		unconnected	—
U5.24	GND		unconnected	—
U5.25	EP		unconnected	—
USB1.1	GND		unconnected	—
USB1.2	COM		unconnected	—
USB1.3	GND		unconnected	—
USB1.4	LB		unconnected	—
USB1.5	GND		unconnected	—
USB1.6	HB		unconnected	—
X5.1	OSC1	C3XTAL_40K_P	connected	—
X5.2	2	GND	connected	—
X5.3	OSC2	C3XTAL_40K_N	connected	—
X5.4	4	GND	connected	—
X6.1	OSC1		unconnected	—
X6.2	GND		unconnected	—
X6.3	OSC2		unconnected	—
X6.4	GND		unconnected	—
X9.1	OSC1		unconnected	—
X9.2	GND		unconnected	—
X9.3	OSC2		unconnected	—
X9.4	GND		unconnected	—

已确认的更正

- U3 19–24 脚为封装内 Flash 信号，外部 NC；已保存重开验证。
- GPIO18/19 是 USB D−/D+；UART0 默认 RX/TX 为 GPIO20/21。主控与 ESP32 的现有 USART 网需要按固件映射核实。
- ESP32-C3 的 microSD 使用 SPI 方案，不能沿用原稿的 GPIO18–23 四位 SDMMC 描述。
- CH334P 使用 12MHz 晶振；器件没有 RREF 脚。
- USB1/2/3 的 DPX205850DT-4055A1 是射频双工器，不是 USB ESD 器件。
- 原脚名推断建议已移至 baseline/documents，不能作为接线依据；尤其 VDD_PA、VR_PA 等脚不能仅按名字并入 3V3。

3. 接线任务表

接线任务与当前证据

> 状态：incomplete。当前统计来自保留快照；脚本不会读取或修改 EDA。

设计 · 全部物理引脚 · 执行报告

ID	任务	状态
T0	基线与审计	已采集；发现电源桥接，不能视为电气通过
T1	2–6S / USB 电源	需求确定；U11 高压降压及保护重设计未完成，U4 按低压域单独核算
T2	射频前端	目标确定：ELRS 1W，nRF24/A7105 500mW，ESP-NOW 无大功率要求；选型未完成
T3	重复器件	X1/X7 尚未删除，待时钟用途核实
T4	资料核实	CH334P 12MHz/无 RREF 已更正；取得 RTL8812AU 参考图，BOM 和射频器件仍不完整
T5	POWER	未完成：C18:2 与 C19:1 的 BATTERY/BATTERY_5V 桥接已确认；严格修复预演拒绝
T6	MCU / USB	未完成：Hub 上下行隔离、GPIO 分配及外围待设计
T7	VTX	未完成：参考图已留档，电源/USB/时钟/射频链未落图
T8	LoRa 控制	部分：U3 19–24 外部 NC 已保存重开验证；其余总线和外围待接。**ESP-NOW 模式硬件需求已定稿(DESIGN §2.2,参考 zvldz/ESP32-UART-Bridge 架构):需新增绑定按键(GPIO0 候选)+ 状态 LED(空闲 GPIO 候选)两器件,走变更清单**
T9	RF 前端	未完成：前端与收发互锁待设计
T10	全图交付	未通过；不能将官方 fatal/error=0 等同于完成

分页物理引脚状态

页	联网	NC	未连接	未知/冲突
POWER	63	0	1	0
MCU	102	0	80	0
VTX	0	0	137	0
LoRa	45	6	198	0

执行阻塞

- 152/153 个实例缺少可证明的完整器件库身份；U3 的 resolve-lcsc 也未能恢复。保留实例的 compose 重建尚不能形成有效输入。
- POWER 的 C18 桥接修复 dry-run 返回 no safe candidate；禁止关闭严格检查强行接线。
- 参考资料不完整的电路仍需选型与核算，不能用 NC 或单引脚标签网络代替。

依赖顺序：恢复身份与布局工具能力 → 电源/前端及接口定版 → 分模块落图 → 保存重开与全图验收。

4. BOM

大疆多协议高频头 — BOM(153 器件)

> 2026-09-24：153 个器件保留，未执行换件/删件；仅 U3 Flash 引脚增加 NC。该表是现有器件清单，不是可投产 BOM。152 个实例的完整库身份未恢复。USB1/2/3 实际为 RF Diplexer；电源与前端器件待重选，详情见 执行报告 和 资料核实。
>
> 来源:2026-09-23 Phase 0 快照(schematic9 旧 8 页 + PCB8 dump)。 页归属为四页迁移目标页;源页为迁移前旧页;C 号为立创商城编号。

页	位号	型号	立创 C 号	封装	值	旧页
LoRa	C11	CC0402KRX7R9BB471	C106207	C0402	470pF	esp3 sd_P
LoRa	C13	CC0402JRNPO9BN100	C106199	C0402	10pF	esp3 sd_P
LoRa	C31	CC0402JRNPO9BN101	C106200	C0402	100pF	rf-pa
LoRa	C32	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	rf-pa
LoRa	C33	CC0402JRNPO9BN101	C106200	C0402	100pF	rf-sw
LoRa	C35	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	rf-sw
LoRa	C36	CC0402JRNPO9BN100	C106199	C0402	10pF	rf-rx
LoRa	C49	CC0402JRNPO9BN101	C106200	C0402	100pF	rf-sw
LoRa	C51	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	rf-sw
LoRa	C58	CL05B104KB54PNC	C307331	C0402	100nF	rf-rx
LoRa	C61	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	rf-rx
LoRa	C62	CGA0805X5R226M250MT	C6119899	C0805	22uF	rf-pa
LoRa	C63	CL05B104KB54PNC	C307331	C0402	100nF	rf-pa
LoRa	C64	CL05B104KB54PNC	C307331	C0402	100nF	rf-sw
LoRa	C65	CL05B104KB54PNC	C307331	C0402	100nF	esp3 sd_P
LoRa	C67	FCC0402N330J500AT	C5137486	C0402	33pF	rf-sw
LoRa	C68	FCC0402N330J500AT	C5137486	C0402	33pF	rf-sw
LoRa	CN1	HX PZ2.0-1x4P ZZ	C22438287	HDR-TH_4P-P2.00-V-M-3	HX PZ2.0-1x4P ZZ	esp3 sd_P
LoRa	CN3	MUP-M627-1	C5205362	MICRO-SD-SMD_MUP-M627-1	MUP-M627-1	esp3 sd_P
LoRa	L18	LQG15HS8N2J02D	C98003	L0402	8.2nH	rf-rx
LoRa	L19	SDCL1005C3N3STDF	C16526	L0402	3.3nH	rf-rx
LoRa	L2	BLM15AG700SN1D	C85814	L0402	BLM15AG700SN1D	rf-rx
LoRa	L20	LQG15HS22NJ02D	C12670	L0402	22nH	rf-rx
LoRa	L23	APS252012M3R3F	C47327425	IND-SMD_L2.5-W2.0_APS252012M3R3F	3.3uH	rf-rx

页	位号	型号	立创 C 号	封装	值	旧页
LoRa	L24	BLM15AG601SN1D	C76884	L0402	BLM15AG601SN1D	rf-pa
LoRa	L25	LQG15HS8N2J02D	C98003	L0402	8.2nH	rf-sw
LoRa	L26	SDCL1005C3N3STDF	C16526	L0402	3.3nH	rf-sw
LoRa	L8	SLFB15-2R450G-02TF	C96749	FILTER-SMD_SLFB15-2R450G-02TF	SLFB15-2R450G-02TF	rf-pa
LoRa	Q2	AO3401A	C15127	SOT-23_L2.9-W1.3-P1.90-LS2.4-BR	AO3401A	rf-pa
LoRa	R21	FRC0402F2202TS	C2909337	R0402	22kΩ	rf-rx_
LoRa	R24	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R25	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R26	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R27	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R28	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R29	FRC0402F1002TS	C2906861	R0402	10kΩ	esp32 sd_P
LoRa	R30	FRC0402F0000TS	C2906858	R0402	0Ω	rf-sw
LoRa	R31	FRC0402F0000TS	C2906858	R0402	0Ω	rf-sw
LoRa	R32	FRC0402F1001TS	C2906864	R0402	1kΩ	rf-pa
LoRa	R33	FRC0402F1002TS	C2906861	R0402	10kΩ	rf-pa
LoRa	U15	ABDFTCXO-52.000MHz-L-2-T5	C568572	OSC-SMD_5P-L14.2-W9.2-BL_ABDFVCTCXO-26.000MH	52MHz	rf-rx_
LoRa	U17	AT2401C	C783588	QFN-16_L3.0-W3.0-P0.50-BL-EP1.7	AT2401C	rf-pa
LoRa	U19	NRF24L01P-R	C8791	QFN-20_L4.0-W4.0-P0.50-BL-EP2.7	无线收发 2.4G(已换绑)	rf-rx_
LoRa	U27	A7105	C126376	QFN-20_L4.0-W4.0-P0.50-BL-EP2.2	A7105	rf-sw
LoRa	U29	SKY13414-485LF	C255353	QFN-14_L2.0-W2.0-P0.45-BL-EP	SKY13414-485LF	rf-sw
LoRa	U3	ESP32-C3FH4	C2858491	QFN-32_L5.0-W5.0-P0.50-TL-EP3.7	ESP32-C3FH4	esp32 sd_P
LoRa	U5	SX1281IMLTRT	C2151551	QFN-24_L4.0-W4.0-P0.50-TL-EP2.6	SX1281IMLTRT	rf-rx_
LoRa	USB1	DPX205850DT-4055A1	C2847617	FILTER-SMD_DPX165950DT-8126A1	DPX205850DT-4055A1	rf-pa
LoRa	X5	Q22FA12800101	C255898	CRYSTAL-SMD_4P-L2.0-W1.6-BL	40MHz	esp32 sd_P
LoRa	X6	X322516MLB4SI	C13738	CRYSTAL-SMD_4P-L3.2-W2.5-BL	16MHz	rf-rx_
LoRa	X9	ABM8AIG-12.000MHZ-12-D2Z-T	C1986775	CRYSTAL-SMD_4P-L3.2-W2.5-BL	12MHz	rf-sw

页	位号	型号	立创 C 号	封装	值	旧页
MCU	C1	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C10	CGA0603X5R105K160JT	C6119849	C0603	1uF	mcu_
MCU	C12	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C14	CC0402JRNPO9BN100	C106199	C0402	10pF	mcu_
MCU	C16	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C2	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C26	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C3	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C4	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C40	CC0402JRNPO9BN100	C106199	C0402	10pF	mcu_
MCU	C41	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C47	CC0402JRNPO9BN101	C106200	C0402	100pF	mcu_
MCU	C5	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C52	CC0402JRNPO9BN101	C106200	C0402	100pF	mcu_
MCU	C55	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C56	CC0402BRNPO9BN1R5	C282238	C0402	1.5pF	mcu_
MCU	C57	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	mcu_
MCU	C59	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C6	CL10A475KO8NNNC	C19666	C0603	4.7uF	mcu_
MCU	C60	CGA0805X5R226M250MT	C6119899	C0805	22uF	mcu_
MCU	C7	CL05B104KB54PNC	C307331	C0402	100nF	mcu_
MCU	C8	CC0402JRNPO9BN100	C106199	C0402	10pF	mcu_
MCU	CN2	HX PZ2.0-1x4P ZZ	C22438287	HDR-TH_4P-P2.00-V-M-3	HX PZ2.0-1x4P ZZ	mcu_
MCU	CN4	HX PZ2.0-1x4P ZZ	C22438287	HDR-TH_4P-P2.00-V-M-3	HX PZ2.0-1x4P ZZ	stagi
MCU	L21	LQG15HS22NJ02D	C12670	L0402	22nH	mcu_
MCU	L22	APS252012M3R3F	C47327425	IND-SMD_L2.5-W2.0_APS252012M3R3F	3.3uH	mcu_
MCU	Q1	AO3401A	C15127	SOT-23_L2.9-W1.3-P1.90-LS2.4-BR	AO3401A	mcu_
MCU	R1	FRC0402F1002TS	C2906861	R0402	10kΩ	mcu_
MCU	R14	FRC0402F1001TS	C2906864	R0402	1kΩ	mcu_
MCU	R16	FRC0402F1001TS	C2906864	R0402	1kΩ	stagi
MCU	R17	FRC0402F1002TS	C2906861	R0402	10kΩ	stagi
MCU	R19	FRC0402F1001TS	C2906864	R0402	1kΩ	mcu_
MCU	R2	FRC0402F1002TS	C2906861	R0402	10kΩ	stagi
MCU	R20	FRC0402F1002TS	C2906861	R0402	10kΩ	mcu_
MCU	R22	FRC0402F1003TS	C2906859	R0402	100kΩ	mcu_
MCU	R6	FRC0402F1002TS	C2906861	R0402	10kΩ	mcu_
MCU	R8	FRC0402F1001TS	C2906864	R0402	1kΩ	stagi
MCU	TP1	?	Test-Point.1	Test-Point-0.5mm		mcu_
MCU	U2	N32H473CCU7	C49191475		N32H473CCU7	stagi

页	位号	型号	立创 C 号	封装	值	旧页
				QFN-48_L7.0-W7.0-P0.50-BL-EP5.6		
MCU	USB2	DPX205850DT-4055A1	C2847617	FILTER-SMD_DPX165950DT-8126A1	DPX205850DT-4055A1	stagi
MCU	USB3	DPX205850DT-4055A1	C2847617	FILTER-SMD_DPX165950DT-8126A1	DPX205850DT-4055A1	mcu_
MCU	X1	SX3B8.000F1010F30	C7302001	CRYSTAL-SMD_4P-L3.2-W2.5-BL	8MHz	stagi
MCU	X2	SX3B8.000F1010F30	C7302001	CRYSTAL-SMD_4P-L3.2-W2.5-BL	8MHz	mcu_
MCU	X7	40MHZ/3225/12PF/10PPM	C47089401	CRYSTAL-SMD_4P-L3.2-W2.5-BL	40MHz	mcu_
POWER	C15	CL05B104KB54PNC	C307331	C0402	100nF	powe io_P1
POWER	C18	CGA0805X5R226M250MT	C6119899	C0805	22uF	powe io_P1
POWER	C19	CGA0805X5R226M250MT	C6119899	C0805	22uF	powe io_P1
POWER	C20	CC0402JRNPO9BN220	C106203	C0402	22pF	powe io_P1
POWER	C27	CC0402JRNPO9BN220	C106203	C0402	22pF	powe io_P1
POWER	C29	CGA0805X5R226M250MT	CGA0805X5R226M250MT.1	C0805	22uF	powe io_P1
POWER	C30	CGA0805X5R226M250MT	C6119899	C0805	22uF	powe io_P1
POWER	C9	CL05B104KB54PNC	C307331	C0402	100nF	powe io_P1
POWER	L10	APS252012M3R3F	C47327425	IND-SMD_L2.5-W2.0_APS252012M3R3F	3.3uH	powe io_P1
POWER	L9	APS252012M3R3F	C47327425	IND-SMD_L2.5-W2.0_APS252012M3R3F	3.3uH	powe io_P1
POWER	R10	HRC0402F9102DNTO	C54920709	R0402	91kΩ	powe io_P1
POWER	R11	FRC0402F1003TS	FRC0402F1003TS.1	R0402	100kΩ	powe io_P1
POWER	R12	ERJ2RKF2433X	ERJ2RKF2433X.1	R0402	243kΩ	powe io_P1
POWER	R15	FRC0402F1002TS	FRC0402F1002TS.1	R0402	10kΩ	powe io_P1
POWER	R3	FRC0402F0000TS	FRC0402F0000TS.1	R0402	0Ω	powe io_P1
POWER	R4	FRC0402F0000TS	FRC0402F0000TS.1	R0402	0Ω	powe io_P1
POWER	R5	FRC0402F2202TS	C2909337	R0402	22kΩ	powe io_P1
POWER	R7	FRC0402F1003TS	C2906859	R0402	100kΩ	powe io_P1

页	位号	型号	立创 C 号	封装	值	旧页
POWER	R9	FRC0402F1202TS	C2909316	R0402	12kΩ	powe io_P1
POWER	U1	ZX-SH1.0-4PWT	ZX-SH1.0-4PWT.1	CONN-SMD_4P-P1.00_MEGASTAR_ZX-SH1.0-4PWT	ZX-SH1.0-4PWT	powe io_P1
POWER	U11	JW5357MSOTB#TR	C5197556	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL	JW5357MSOTB#TR	powe io_P1
POWER	U13	TPS2117DRLR	TPS2117DRLR_C22399676.1	SOT-583-8_L2.1-W1.2-P0.50-LS1.6-BL	TPS2117DRLR	powe io_P1
POWER	U4	JW5357MSOTB#TR	C5197556	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL	JW5357MSOTB#TR	powe io_P1
VTX	C17	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C21	CGA0603X5R105K160JT	C6119849	C0603	1uF	stagi
VTX	C22	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C23	CC0402JRNPO9BN100	C106199	C0402	10pF	stagi
VTX	C24	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C25	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C28	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C34	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C37	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C38	CC0402JRNPO9BN100	C106199	C0402	10pF	stagi
VTX	C39	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C42	CC0402JRNPO9BN101	C106200	C0402	100pF	stagi
VTX	C43	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C44	CC0402JRNPO9BN101	C106200	C0402	100pF	stagi
VTX	C45	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C46	CC0402BRNPO9BN1R5	C282238	C0402	1.5pF	stagi
VTX	C48	CC0402BRNPO9BN8R2	C327205	C0402	8.2pF	stagi
VTX	C50	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C53	CL10A475KO8NNNC	C19666	C0603	4.7uF	stagi
VTX	C54	CGA0805X5R226M250MT	C6119899	C0805	22uF	stagi
VTX	C66	CL05B104KB54PNC	C307331	C0402	100nF	stagi
VTX	C69	CC0402JRNPO9BN100	C106199	C0402	10pF	stagi
VTX	L1	LQG15HS22NJ02D	C12670	L0402	22nH	stagi
VTX	L3	APS252012M3R3F	C47327425	IND-SMD_L2.5-W2.0_APS252012M3R3F	3.3uH	stagi
VTX	Q3	AO3401A	C15127	SOT-23_L2.9-W1.3-P1.90-LS2.4-BR	AO3401A	stagi
VTX	R18	FRC0402F1003TS	C2906859	R0402	100kΩ	stagi
VTX	R23	FRC0402F1002TS	C2906861	R0402	10kΩ	vtx_P
VTX	R34	FRC0402F1002TS	C2906861	R0402	10kΩ	stagi
VTX	TP2	?	Test-Point.1	Test-Point-0.5mm		stagi
VTX	U14	ZX-SH1.0-4PWT	C7430446		ZX-SH1.0-4PWT	vtx_P

页	位号	型号	立创 C 号	封装	值	旧页
				CONN-SMD_4P-P1.00_MEGASTAR_ZX-SH1.0-4PWT		
VTX	U20	CH334P	C5373042	QFN-16_L3.0-W3.0-P0.50-BL-EP1.7	CH334P	vtx_P
VTX	U21	RTL8812AU-CG	C2928090	QFN-76_L9.0-W9.0-P0.40-TL-EP	RTL8812AU-CG	stagi
VTX	U8	RCLAMP0521T-ES	C5180263	DFN1006-2L-BI	RCLAMP0521T-ES	vtx_P
VTX	U9	RCLAMP0521T-ES	C5180263	DFN1006-2L-BI	RCLAMP0521T-ES	vtx_P
VTX	X3	40MHZ/3225/12PF/10PPM	C47089401	CRYSTAL-SMD_4P-L3.2-W2.5-BL	40MHz	stagi